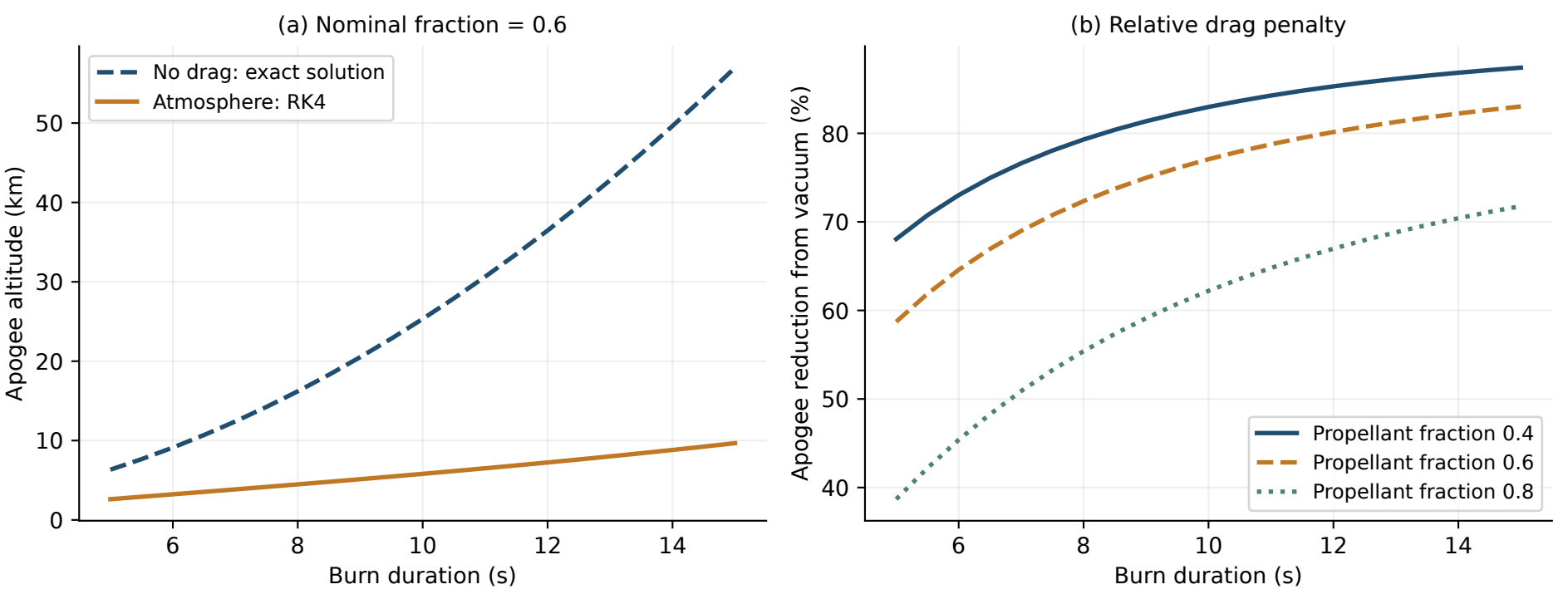
(a) Nominal fraction = 0.6
No drag: exact solution
Atmosphere: RK4
Apogee altitude (km)
0
10
20
30
40
50
6
8
10
12
14
Burn duration (s)
(b) Relative drag penalty
Apogee reduction from vacuum (%)
40
50
60
70
80
Propellant fraction 0.4
Propellant fraction 0.6
Propellant fraction 0.8
6
8
10
12
14
Burn duration (s)